

TEST REPORT

DER Scheduling IEC 61850 Conformance Test

2024-07-26

Test Details

test setup	For this test, both the Alliander DER Scheduler and the test routines are running on Ubuntu 20 LTS. The unit tests initiate the execution on IEC 61850 schedules by utilizing a IEC 61850 client written in Java ("IEC61850bean"). The tests are based on the IEC 61850 spec, especially IEC 61850-90-10:2017 and "Lastenheft Steuerbox: Funktionale und konstruktive Merkmale v1.3".
requirements	Requirements, e.g. 'S01' are references to identifiers to the requirements on Alliander DER Scheduling on github. For 'S01' means "A set of 10 schedules per logical node must be supported". For a full overview, visit: https://github.com/alliander-opensource/der-scheduling/blob/main/REQUIREMENTS.md
tested version	https://github.com/alliander-opensource/der-scheduling/releases/tag/conformance-tests-v1.1
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Test Session Summary

Start: Fri Jul 26 13:55:33 2024

End: Fri Jul 26 14:09:32 2024

Duration: 13m 59s

passed: 88 (100.00%)

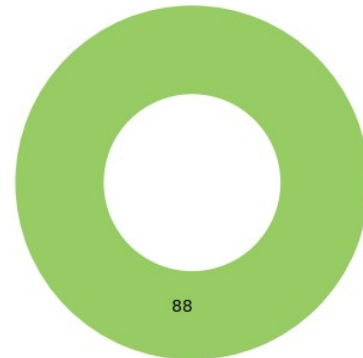
skipped: 0 (0.00%)

broken: 0 (0.00%)

failed: 0 (0.00%)

unknown: 0 (0.00%)

 passed



ActStrTmIsUpdatedProperly running active power schedules [2] [passed]

Duration	45s
Requirements	LN03
Description	Test if optional node ActStrTm is present it should behave like defined in IEC 61850-90-10:2017, table 7 (page 26)

ActStrTmIsUpdatedProperly running max power schedules [0] [passed]

Duration	45s
Requirements	LN03
Description	Test if optional node ActStrTm is present it should behave like defined in IEC 61850-90-10:2017, table 7 (page 26)

ActStrTmIsUpdatedProperly running on/off schedules [1] [passed]

Duration	45s
Requirements	LN03
Description	Test if optional node ActStrTm is present it should behave like defined in IEC 61850-90-10:2017, table 7 (page 26)

DsaReq_operating running active power schedules [1] [passed]

Duration	40s
Requirements	LN03
Description	Test that DsaReq behaves as described in IEC 61850-90-10:2017, table 7 (page 26)

DsaReq_operating running max power schedules [0] [passed]

Duration	40s
Requirements	LN03
Description	Test that DsaReq behaves as described in IEC 61850-90-10:2017, table 7 (page 26)

DsaReq_operating running on/off schedules [2] [passed]

Duration	40s
Requirements	LN03
Description	Test that DsaReq behaves as described in IEC 61850-90-10:2017, table 7 (page 26)

EnaReq_operating running active power schedules [1] [passed]

Duration	20s
Requirements	LN03
Description	Test that EnaReq holds a reasonable error code after provoking errors whilst enabling schedules. See IEC 61850-90-10:2017, table 7 (page 26)

EnaReq_operating running max power schedules [2] [passed]

Duration	20s
Requirements	LN03
Description	Test that EnaReq holds a reasonable error code after provoking errors whilst enabling schedules. See IEC 61850-90-10:2017, table 7 (page 26)

EnaReq_operating running on/off schedules [0] [passed]

Duration	20s
Requirements	LN03
Description	Test that EnaReq holds a reasonable error code after provoking errors whilst enabling schedules. See IEC 61850-90-10:2017, table 7 (page 26)

IEC 61850 Protocol Library version check [passed]**Duration** 22ms**Description** Checks installed library version of libiec61850. Expects v1.5.2.**NumEntr_range running active power schedules [1] [passed]****Duration** 217ms**Requirements** LN03**Description** Test that NumEntr can only be set to values > 0 and values $\leq$ the number of instantiated Val[ASG|ING|SPG|ENG]'s as stated in IEC 61850-90-10:2017, table 7 (page 26)**NumEntr_range running max power schedules [0] [passed]****Duration** 219ms**Requirements** LN03**Description** Test that NumEntr can only be set to values > 0 and values $\leq$ the number of instantiated Val[ASG|ING|SPG|ENG]'s as stated in IEC 61850-90-10:2017, table 7 (page 26)**NumEntr_range running on/off schedules [2] [passed]****Duration** 240ms

Requirements	LN03
Description	Test that NumEntr can only be set to values > 0 and values <= the number of instantiated Val[ASG ING SPG ENG]'s as stated in IEC 61850-90-10:2017, table 7 (page 26)

NxtStrTmlsUpdatedProperly running active power schedules [1] [passed]

Duration	23ms
Requirements	LN03
Description	Test if NxtStrTm is mandatory and should behave like defined in IEC 61850-90-10:2017, table 7 (page 26)

NxtStrTmlsUpdatedProperly running max power schedules [0] [passed]

Duration	24ms
Requirements	LN03
Description	Test if NxtStrTm is mandatory and should behave like defined in IEC 61850-90-10:2017, table 7 (page 26)

NxtStrTmIsUpdatedProperly running on/off schedules [2] [passed]

Duration	24ms
Requirements	LN03
Description	Test if NxtStrTm is mandatory and should behave like defined in IEC 61850-90-10:2017, table 7 (page 26)

SchdEnaErrHoldsMISSING_VALID_SCHDINTVcorrectly running active power schedules [1] [passed]

Duration	2s
Requirements	LN03 S08
Description	Test that SchdEnaErr holds MISSING_VALID_SCHDINTV when writing invalid value to SchdIntv.setVal (see IEC 61850-90-10:2017, table 7 (page 26))

SchdEnaErrHoldsMISSING_VALID_SCHDINTVcorrectly running max power schedules [2] [passed]

Duration	2s
Requirements	LN03 S08

Description	Test that SchdEnaErr holds MISSING_VALID_SCHDINTV when writing invalid value to SchdIntv.setVal (see IEC 61850-90-10:2017, table 7 (page 26))
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SchdEnaErrHoldsMISSING_VALID_SCHDINTVcorrectly running on/off schedules [0] [passed]

Duration	2s
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Requirements	LN03
	S08

Description	Test that SchdEnaErr holds MISSING_VALID_SCHDINTV when writing invalid value to SchdIntv.setVal (see IEC 61850-90-10:2017, table 7 (page 26))
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SchdEnaErrHoldsMISSING_VALID_SCHEDULE_VALUEscorrectly running active power schedules [1] [passed]

Duration	2s
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Requirements	LN03
	S08

Description	Tests that SchdEnaErr holds the error code MISSING_VALID_SCHEDULE_VALUE when invalid values are written, for (Max)Power Schedules (see IEC 61850-90-10:2017, table 7 (page 26))
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SchdEnaErrHoldsMISSING_VALID_SCHEDULE_VALUEscorrectly running max power schedules [0] [passed]

Duration	2s
Requirements	LN03 S08
Description	Tests that SchdEnaErr holds the error code MISSING_VALID_SCHEDULE_VALUE when invalid values are written, for (Max)Power Schedules (see IEC 61850-90-10:2017, table 7 (page 26))

SchdEntrIsUpdatedWithCurrentlyRunningScheduleIfPresent running active power schedules [2] [passed]

Duration	40s
Requirements	LN03
Description	Test that SchdEntr is present and is updated by the currently running schedule as described in IEC 61850-90-10:2017, table 7 (page 26)

SchdEntrIsUpdatedWithCurrentlyRunningScheduleIfPresent running max power schedules [1] [passed]

Duration	40s
Requirements	LN03
Description	Test that SchdEntr is present and is updated by the currently running schedule as described in IEC 61850-90-10:2017, table 7 (page 26)

SchdEntrIsUpdatedWithCurrentlyRunningScheduleIfPresent running on/off schedules [0] [passed]

Duration	40s
Requirements	LN03
Description	Test that SchdEntr is present and is updated by the currently running schedule as described in IEC 61850-90-10:2017, table 7 (page 26)

ValEnsIsUpdatedWithCurrentlyRunningScheduleIfPresent running active power schedules [2] [passed]

Duration	3ms
Requirements	LN03
Description	Test that ValENS is not available

ValEnslsUpdatedWithCurrentlyRunningScheduleIfPresent running max power schedules [0] [passed]

Duration	3ms
Requirements	LN03
Description	Test that ValENS is not available

ValEnslsUpdatedWithCurrentlyRunningScheduleIfPresent running on/off schedules [1] [passed]

Duration	2ms
Requirements	LN03
Description	Test that ValENS is not available

ValINSlsUpdatedWithCurrentlyRunningScheduleIfPresent running active power schedules [0] [passed]

Duration	3ms
Requirements	LN03
Description	Test that ValINS is not available

ValNSIsUpdatedWithCurrentlyRunningScheduleIfPresent running max power schedules [1] [passed]

Duration	2ms
Requirements	LN03
Description	Test that ValINS is not available

ValNSIsUpdatedWithCurrentlyRunningScheduleIfPresent running on/off schedules [2] [passed]

Duration	2ms
Requirements	LN03
Description	Test that ValINS is not available

ValSpsOrValMvIsUpdatedWithCurrentlyRunningSchedule running active power schedules [1] [passed]

Duration	39s
Requirements	LN03
Description	Test ValSPS/ValMv behaves as defined in IEC 61850-90-10:2017, table 7 (page 26)

ValSpsOrValMvIsUpdatedWithCurrentlyRunningSchedule running max power schedules [2] [passed]

Duration	39s
Requirements	LN03
Description	Test ValSPS/ValMv behaves as defined in IEC 61850-90-10:2017, table 7 (page 26)

ValSpsOrValMvIsUpdatedWithCurrentlyRunningSchedule running on/off schedules [0] [passed]

Duration	39s
Requirements	LN03
Description	Test ValSPS/ValMv behaves as defined in IEC 61850-90-10:2017, table 7 (page 26)

absolutePowerValueSchedulesAreSupported() [passed]

Duration	5ms
Requirements	S05a

activeControllerIsUpdated running active power schedules [0] [passed]

Duration 3s

Requirements LN03
LN02c
LN02b
LN02a

activeControllerIsUpdated running max power schedules [1] [passed]

Duration 3s

Requirements LN03
LN02c
LN02b
LN02a

activeControllerIsUpdated running on/off schedules [2] [passed]

Duration 3s

Requirements LN03
LN02c

LN02b

LN02a

**activeControllerIsUpdatedWithScheduleOfHighestPrio running
active power schedules [2] [passed]**

Duration 2s

Requirements LN03
E01

**activeControllerIsUpdatedWithScheduleOfHighestPrio running
max power schedules [0] [passed]**

Duration 2s

Requirements LN03
E01

**activeControllerIsUpdatedWithScheduleOfHighestPrio running
on/off schedules [1] [passed]**

Duration 2s

Requirements LN03
E01

**allExpectedSchedulesExist running active power schedules [0]
[passed]**

Duration 5ms

Requirements LN02c
LN02b
LN02a

**allExpectedSchedulesExist running max power schedules [2]
[passed]**

Duration 5ms

Requirements LN02c
LN02b
LN02a

allExpectedSchedulesExist running on/off schedules [1] [passed]

Duration 4ms

Requirements LN02c
LN02b

LN02a

checkSubnodes running active power schedules [1] [passed]

Duration	14ms
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Requirements	LN02
	LN01
	LN03

checkSubnodes running max power schedules [2] [passed]

Duration	3ms
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Requirements	LN02
	LN01
	LN03

checkSubnodes running on/off schedules [0] [passed]

Duration	4ms
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Requirements	LN02
	LN01
	LN03

hasRequiredSubNodes running active power schedules [2] [passed]

Duration	17ms
Requirements	LN01 LN03
Description	Test if the scheduler has the required nodes with correct types as defined in IEC 61850-90-10:2017, table 7 (page 26)

hasRequiredSubNodes running max power schedules [0] [passed]

Duration	5ms
Requirements	LN01 LN03
Description	Test if the scheduler has the required nodes with correct types as defined in IEC 61850-90-10:2017, table 7 (page 26)

hasRequiredSubNodes running on/off schedules [1] [passed]

Duration	4ms
Requirements	LN01 LN03
Description	Test if the scheduler has the required nodes with correct types as defined in IEC 61850-90-10:2017, table 7 (page 26)

maxPowerValueSchedulesAreSupported() [passed]

Duration	5ms
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Requirements	S05b
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onOffSchedulesAreSupported() [passed]

Duration	4ms
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Requirements	S05c
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**reserveScheduleSupports100values running active power
schedules [2] [passed]**

Duration	27ms
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Requirements	S12
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**reserveScheduleSupports100values running max power
schedules [0] [passed]**

Duration	7ms
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Requirements	S12
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reserveScheduleSupports100values running on/off schedules [1]
[passed]

Duration 8ms

Requirements S12

reserveSchedulesCannotBeDeactivated running active power
schedules [1] [passed]

Duration 4ms

Requirements S13

reserveSchedulesCannotBeDeactivated running max power
schedules [2] [passed]

Duration 4ms

Requirements S13

reserveSchedulesCannotBeDeactivated running on/off schedules [0] [passed]

Duration 3ms

Requirements S13

reserveSchedulesExist running active power schedules [0] [passed]

Duration 5ms

Requirements S04

reserveSchedulesExist running max power schedules [1] [passed]

Duration 4ms

Requirements S04

reserveSchedulesExist running on/off schedules [2] [passed]

Duration 3ms

Requirements S04

**reserveSchedulesHaveFixedPriority running active power
schedules [0] [passed]**

Duration 2ms

Requirements S14

**reserveSchedulesHaveFixedPriority running max power
schedules [2] [passed]**

Duration 3ms

Requirements S14

**reserveSchedulesHaveFixedPriority running on/off schedules [1]
[passed]**

Duration 3ms

Requirements S14

**reserveSchedulesHaveFixedStart running active power
schedules [2] [passed]**

Duration 4ms

Requirements S15

**reserveSchedulesHaveFixedStart running max power schedules
[1] [passed]**

Duration 2ms

Requirements S15

**reserveSchedulesHaveFixedStart running on/off schedules [0]
[passed]**

Duration 3ms

Requirements S15

**schdEnaErr_HoldsMISSING_VALID_NUMENTRcorrectly running
active power schedules [1] [passed]**

Duration 2s

Requirements	LN03 S08
Description	Test that SchdEnaErr holds MISSING_VALID_NUMENTR when writing invalid value to NumEntr.setVal (see IEC 61850-90-10:2017, table 7 (page 26))

schdEnaErr_HoldsMISSING_VALID_NUMENTRcorrectly running max power schedules [2] [passed]

Duration	2s
Requirements	LN03 S08
Description	Test that SchdEnaErr holds MISSING_VALID_NUMENTR when writing invalid value to NumEntr.setVal (see IEC 61850-90-10:2017, table 7 (page 26))

schdEnaErr_HoldsMISSING_VALID_NUMENTRcorrectly running on/off schedules [0] [passed]

Duration	2s
Requirements	LN03 S08
Description	Test that SchdEnaErr holds MISSING_VALID_NUMENTR when writing invalid value to NumEntr.setVal (see IEC 61850-90-10:2017, table 7 (page 26))

scheduleSupports100values running active power schedules [0]
[passed]

Duration 16ms

Requirements S10

scheduleSupports100values running max power schedules [1]
[passed]

Duration 7ms

Requirements S10

scheduleSupports100values running on/off schedules [2]
[passed]

Duration 7ms

Requirements S10

**scheduleSupportsTimebasedScheduling running active power
schedules [2] [passed]**

Duration 5ms

Requirements S02

**scheduleSupportsTimebasedScheduling running max power
schedules [0] [passed]**

Duration 4ms

Requirements S02

**scheduleSupportsTimebasedScheduling running on/off
schedules [1] [passed]**

Duration 4ms

Requirements S02

**tenSchedulesAreSupportedPerType running active power
schedules [1] [passed]**

Duration 5ms

Requirements	S01
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**tenSchedulesAreSupportedPerType running max power
schedules [0] [passed]**

Duration	5ms
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Requirements	S01
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**tenSchedulesAreSupportedPerType running on/off schedules [2]
[passed]**

Duration	4ms
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Requirements	S01
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**testSamePriosDifferentStartOnOffSchedule running on/off
schedules [passed]**

Duration	11s
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Description	IEC61850-90-10 ed 2017 Schedule Controller Definitions, section 5.5.3
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testSamePriosDifferentStartPowerSchedules running active power schedules [1] [passed]

Duration	48s
Description	IEC61850-90-10 ed 2017 Schedule Controller Definitions, section 5.5.3

testSamePriosDifferentStartPowerSchedules running max power schedules [0] [passed]

Duration	48s
Description	IEC61850-90-10 ed 2017 Schedule Controller Definitions, section 5.5.3

test_prioritiesOnOffSchedules running on/off schedules [passed]

Duration	17s
Requirements	S05c E01 S02 E02

test_prioritiesPowerSchedules running active power schedules [1] [passed]

Duration	36s
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Requirements	LN01
	S05c
	E01
	S02
	E02
	S09

test_prioritiesPowerSchedules running max power schedules [0] [passed]

Duration	36s
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Requirements	LN01
	S05c
	E01
	S02
	E02
	S09

**test_samePrioAndStartFloatSchedule running active power
schedules [0] [passed]**

Duration 16s

Requirements S17

**test_samePrioAndStartFloatSchedule running max power
schedules [1] [passed]**

Duration 16s

Requirements S17

**test_samePrioAndStartOnOffSchedule running on/off schedules
[passed]**

Duration 16s

Requirements S17

threeReserveSchedulesExist() [passed]

Duration 5ms

Requirements S11

